

1 Deligne's image comparison inside the Split-Zero source complexes

This sidebar integrates the delivered specialization-curvature, exponential-comparison, and logarithmic-constituent-control notes. Their logarithmic cohomology and constituent-coupling calculations are reproduced with complete proofs and explicitly joined to the existing Split-Zero complexes; they are attributed to those web contributions. The sidebar supplies the maps connecting the compact-support comparison, its dual, the Split-Zero internal quotient, and the compactified curve of arithmetic representatives. It also calculates the logarithmic de Rham complex of that curve. In particular, it identifies which cohomology groups the exceptional directions occupy and the divisor twist in their support-compatible dual.

The source reference is P. Deligne, *La conjecture de Weil II*, Publications Mathématiques de l'IHÉS **52** (1980), 137–252, https://www.numdam.org/item/PMIHES_1980__52__137_0/. The specific weight comparison is in (3.3.4)–(3.3.6), printed page 206; the preceding geometric argument is in (3.2.8)–(3.2.13). The source-language S20 records for printed pages 201–207 were read in full, including the strict inequality in (3.2.10) and the squared eigenvalue in (3.2.13). The complete S20 typed transcription is retained separately from this integration. The symbol Q_0 in (DS7)–(DS8) denotes a finite-field cardinality; later Q_0 denotes the explicitly defined quotient Gram, as in the supplied logarithmic note; q denotes the dimension of the actual arithmetic packet.

1.1 The comparison image, its entire cone, and its dual

Let $c : C^\bullet \rightarrow D^\bullet$ be a cochain map of complexes of vector spaces over a field K . All differentials raise degree by one. Write $Z_C^m = \ker d_C^m$, $B_C^m = \operatorname{im} d_C^{m-1}$, and similarly for D . Define

$$K_c^m = \{x \in Z_C^m : c(x) \in B_D^m\}, \quad I_c^m = \operatorname{im}(H^m(c) : H^m(C) \rightarrow H^m(D)). \quad (\text{DS1})$$

There are exact sequences with the displayed maps:

$$\begin{aligned} 0 \longrightarrow K_c^m / B_C^m &\xrightarrow{[x] \mapsto [x]} H^m(C) \xrightarrow{[x] \mapsto [cx]} I_c^m \longrightarrow 0, \\ 0 \longrightarrow I_c^m &\longrightarrow H^m(D) \longrightarrow H^m(D) / I_c^m \longrightarrow 0. \end{aligned} \quad (\text{DS2})$$

Indeed, $B_C^m \subset K_c^m$ because c commutes with the differentials. The kernel of the second arrow in the first sequence is exactly the classes represented by K_c^m , and every element of I_c^m has the indicated preimage. The second sequence is the defining inclusion and quotient. Thus both the lost source classes and the extra target classes are present.

The single complex retaining both sides is

$$\operatorname{Cone}(c)^m = D^m \oplus C^{m+1}, \quad d(y, x) = (d_D y + cx, -d_C x). \quad (\text{DS3})$$

Expansion gives $d^2(y, x) = (d_D^2 y + d_D cx - cd_C x, d_C^2 x) = 0$. The maps $y \mapsto (y, 0)$ and $(y, x) \mapsto x$ give the exact sequence of complexes $0 \rightarrow D \rightarrow \operatorname{Cone}(c) \rightarrow C[1] \rightarrow 0$, where $C[1]^m = C^{m+1}$ has differential $-d_C$. To make the relevant cohomology segment explicit, a cone cocycle obeys $d_C x = 0$ and $cx = -d_D y$, so its projection represents a class in K_c^{m+1} / B_C^{m+1} . Conversely, if $cx = d_D v$, then $(-v, x)$ is a cone cocycle. A projected class is zero precisely when subtracting a cone boundary leaves a representative $(y, 0)$ with $d_D y = 0$. Finally $(y, 0)$ is a cone boundary precisely when $y = d_D u + cv$ with $d_C v = 0$. This proves the exact segment

$$H^m(C) \xrightarrow{H^m(c)} H^m(D) \longrightarrow H^m(\operatorname{Cone}(c)) \longrightarrow H^{m+1}(C) \xrightarrow{H^{m+1}(c)} H^{m+1}(D). \quad (\text{DS4})$$

Every chain action commuting with c acts on (DS1)–(DS4), since it preserves both cycles and boundaries; on the cone its action is the direct sum of the two specified actions.

For a finite-dimensional linear map $f : V \rightarrow W$, its ordinary linear transpose $f^\vee : W^\vee \rightarrow V^\vee$ gives a perfect pairing

$$\operatorname{im} f \times \operatorname{im} f^\vee \longrightarrow K, \quad (f(v), f^\vee \lambda) \longmapsto \lambda(f(v)). \quad (\text{DS5})$$

If v is changed by $\ker f$, the value is unchanged. If λ is changed by $\ker f^\vee$, its restriction to $\operatorname{im} f$ is unchanged. A nonzero $f(v)$ is detected by some functional on W , and a nonzero $f^\vee \lambda$ is detected by some v , proving nondegeneracy on both sides. In particular

$$(\operatorname{im} f)^\vee \simeq \operatorname{im} f^\vee, \quad \ker f^\vee = (\operatorname{im} f)^\perp, \quad \operatorname{im} f^\vee = (\ker f)^\perp. \quad (\text{DS6})$$

For invertible actions T_V, T_W with $fT_V = T_W f$, the dual actions are $(T_V^{-1})^\vee, (T_W^{-1})^\vee$; their induced actions on the two images make the evaluation in (DS5) invariant. For an additive action A and a specified scalar b , the action $bI - A^\vee$ on the dual instead satisfies $\langle Av, \lambda \rangle + \langle v, (bI - A^\vee)\lambda \rangle = b\langle v, \lambda \rangle$. These are two explicitly defined action conventions, with different types of weight parameter.

Deligne's comparison takes

$$C = R\Gamma_c(X, \mathcal{F}), \quad D = R\Gamma(X, \mathcal{F}), \quad c = c_{X, \mathcal{F}}, \quad I_c^m = \operatorname{im} c_m. \quad (\text{DS7})$$

Here $X = X_0 \otimes_{\mathbb{F}_{Q_0}} \overline{\mathbb{F}}_{Q_0}$, X_0 is smooth of pure dimension N and finite type, and $\mathcal{F}_0$ is a smooth $\overline{\mathbb{Q}_\ell}$ -sheaf, $\ell \neq \operatorname{char} \mathbb{F}_{Q_0}$, pure of weight n . In the source theorem the upper weight bound on H_c^m is $n + m$. The dual of $\mathcal{F}$ has weight $-n$, and Poincaré duality gives the exact degree and twist

$$H^m(X, \mathcal{F})^\vee \simeq H_c^{2N-m}(X, \mathcal{F}^\vee)(N). \quad (\text{DS8})$$

The right side has upper weight $-n + (2N - m) - 2N = -n - m$ by the already established compact-support theorem. Dualizing gives the lower weight bound $n + m$ on H^m . Since c_m is Frobenius equivariant, choose a basis first adapted to its kernel in the source and then to its image in the target. The corresponding action matrices are block triangular. Every eigenvalue on I_c^m therefore occurs on both H_c^m and H^m , with its algebraic conjugates. The two bounds force its weight to equal $n + m$. This proves the image step (3.3.6) from the two stated source bounds and proves exactly why (DS7) retains the cone (DS3).

The tensor comparison has an equally explicit algebraic map. For finite-dimensional $f_a : V_a \rightarrow W_a$ ($1 \leq a \leq r$),

$$\operatorname{im} \left(\bigotimes_{a=1}^r f_a \right) = \bigotimes_{a=1}^r \operatorname{im} f_a \subseteq \bigotimes_{a=1}^r W_a. \quad (\text{DS9})$$

The tensor map sends a pure tensor to the pure tensor of its images, so its image is contained in the right side. Conversely these image pure tensors span the right side and each has a preimage; inclusivity into the target follows by extending a basis of each image to a basis of W_a . Thus tensor powers preserve the same comparison image. For complexes, the tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy$; swapping the factors has the sign $(-1)^{\deg x \deg y}$. In characteristic zero a specified symmetry projection is $\frac{1}{r!} \sum_{\pi \in \mathfrak{S}_r} \pi$ with those graded signs. Its idempotence follows because each group element occurs exactly $r!$ times in the double sum. No unspecific tensor eigenvalue is substituted for the eigenvalue on this actual image.

1.2 The internal Split-Zero lift of the same maps

For a complex vector space V , set $S(V) = \{\tau\} \sqcup V^\bullet$. Supported addition and scalar multiplication are the original operations: $v^\bullet + w^\bullet = (v + w)^\bullet$ and $a^\bullet v^\bullet = (av)^\bullet$. External τ is the additive identity and is absorbing for scalar multiplication. The supported zero is $e_V = 0_V^\bullet$. Every linear map $f : V \rightarrow W$ has the canonical lift

$$S(f)(\tau) = \tau, \quad S(f)(v^\bullet) = f(v)^\bullet. \quad (\text{DS10})$$

Checking the two supported inputs and inputs containing τ proves additivity, the scalar law and composition. For $B \subseteq V$, $S(V) \rightarrow S(V/B)$ sends $b^\bullet$ to $e_{V/B}$ for every $b \in B$, and sends τ to τ . Two supported vectors have the same image exactly when their difference lies in B . This is the coequalizer of inclusion $S(B) \rightarrow S(V)$ and the map $b^\bullet \mapsto e_V, \tau \mapsto \tau$: any map equalizing these has equal values on $v^\bullet$ and $(v + b)^\bullet$, by additivity, and hence has a unique factorization through the displayed quotient.

Applying (DS10) to all maps in (DS1)–(DS4) lifts their source kernels, images, quotients and actions. A differential squared is the supported-zero map on represented vectors and still sends τ to τ . The ordinary kernel $\ker H^m(c)$ is carried by the represented fibre of $S(H^m(c))^{-1}(e)$; the preimage of external absence is only $\{\tau\}$. These two fibres are disjoint. Their exact common carrier is $S(\ker H^m(c)) = \{\tau\} \sqcup (\ker H^m(c))^\bullet$, with the inclusion of each of the two fibres. If $p_V : S(V) \rightarrow V$ sends $v^\bullet$ to v and τ to 0, this carrier is precisely $p_V^{-1}(\ker H^m(c))$. Thus the amplitude projection proves the two inclusions while retaining the two distinct receiving zeros. With a join semilattice of supports, carry out the construction in each fibre and use the original transition maps; (DS1)–(DS4) commute with those transitions because their definitions use the commuting cochain maps. No support label is discarded.

The coefficient comparison remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad p_R(a^\bullet) = a, \quad p_R(\tau) = 0. \quad (\text{DS11})$$

It commutes on every supported integer and on τ . In particular the linear/coherent calculation below is the amplitude image of a specified Split-Zero diagram; its supported exceptional kernel remains available before applying p .

1.3 The original source relation produces the curve

Fix the actual cyclic object $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, and the original action $A = M_S$. The monic source spaces $\mathcal{P}_N$, their positive Gram matrices O_N , and their full remainder maps $B_N : \mathcal{P}_N \rightarrow E$ give

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N. \quad (\text{DS12})$$

Take $i \leq j$ with both remainder maps onto, and retain the source inclusion L_{ij} with $B_j L_{ij} = B_i$ and $L_{ij}^* O_j L_{ij} = O_i$. Put

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z \Delta. \quad (\text{DS13})$$

The equalities follow directly: $B_N R_N = I$, $R_j^* O_j \Delta = G_j B_j \Delta = 0$, and expansion of $(L_{ij} R_i)^* O_j (L_{ij} R_i)$ gives $G_j + \Delta^* O_j \Delta$. For the kernel order, the actual source projection $P = L_{ij} O_i^{-1} L_{ij}^* O_j$ obeys $P^2 = P$ and $P^* O_j = O_j P$ by the retained source isometry. Therefore $K_j - K_i = B_j (I - P) O_j^{-1} (I - P)^* B_j^* \succeq 0$; expanding the right side gives $B_j O_j^{-1} B_j^* - B_i O_i^{-1} B_i^*$. Therefore

$$B_j R(z) = I, \quad R(z)^* O_j R(z) = G_j + |z|^2 C. \quad (\text{DS14})$$

The source is the original theta source with its arithmetic coefficient unit and every full jet; the new variable z parametrizes only the comparison and does not replace S .

Let $E_0 = \ker \Delta$, $E_1 = E_0^{\perp_{G_j}}$, $r = \dim E_1$, and let P_0, P_1 be the specified projections. On $X = \mathbb{P}_{\mathbb{C}}^1$ with coordinates $[u : v]$, finite coordinate $z = v/u$ and infinity coordinate $w = u/v$, set $D = \{\infty\}$ and

$$\mathcal{E} = (E_0 \otimes \mathcal{O}_X) \oplus (E_1 \otimes \mathcal{O}_X(-D)). \quad (\text{DS15})$$

The source morphism is

$$\Psi(x_0 + x_1 \otimes (u, v)) = R_j x_0 + u R_j x_1 + v \Delta x_1. \quad (\text{DS16})$$

It is fibrewise injective. At $u \neq 0$, apply B_j and use $E_0 \cap E_1 = 0$; at $u = 0$, apply B_j to force $x_0 = 0$ and then use injectivity of $\Delta|_{E_1}$. Thus the original source Gram defines a smooth positive metric on this bundle, including infinity. The observation $\alpha = B_j \Psi$ is identity in the finite frame and $P_0 + w P_1$ in the infinity frame. Consequently

$$0 \rightarrow \mathcal{E} \xrightarrow{\alpha} E \otimes \mathcal{O}_X \rightarrow i_* E_1 \rightarrow 0, \quad (\text{DS17})$$

where $i : D \hookrightarrow X$. In fact its image lattice is $\mathcal{M} = E_0\mathcal{O}_X + E\mathcal{O}_X(-D)$; at infinity it is $E_0\mathcal{O} + wE\mathcal{O}$.

Tensoring the two locally free terms of (DS17) with the residue field at infinity gives $[E \xrightarrow{P_0} E]$ in degrees $-1, 0$. Its cohomology is E_1 in both degrees. Intrinsically the negative term is $E_1 \otimes \mathcal{O}(-D)|_D$. This calculation is the derived fibre of the skyscraper cokernel, and explicitly retains the Tor term that ordinary right-exact specialization would omit. Dualizing (DS17) gives

$$0 \rightarrow E^*\mathcal{O}_X \rightarrow \mathcal{E}^\vee \rightarrow i_*(E_1^* \otimes \mathcal{O}(D)|_D) \rightarrow 0. \quad (\text{DS18})$$

To check its terminal term, resolve $\mathcal{O}_D$ by $[\mathcal{O}(-D) \rightarrow \mathcal{O}]$ and apply $\mathcal{H}om(-, \mathcal{O})$. The resulting cokernel is $\mathcal{O}(D)|_D$. The two boundary factors pair by evaluation and $\mathcal{O}(-D)|_D \otimes \mathcal{O}(D)|_D \rightarrow \mathbb{C}_D$.

1.4 A calculated compact-support/ordinary de Rham comparison

Transport the trivial connection through α on $X \setminus D$. All connection complexes here are complexes of sheaves of $\mathbb{C}$ -vector spaces: the differentials satisfy the Leibniz rule and are not asserted to be $\mathcal{O}_X$ -linear. In the infinity frame it is

$$\nabla = d + P_1 \frac{dw}{w}, \quad \mathcal{K} = [\mathcal{E} \xrightarrow{\nabla} \mathcal{E} \otimes \Omega_X^1(\log D)] \quad \text{in degrees } 0, 1. \quad (\text{DS19})$$

Indeed $\alpha^{-1}d\alpha = P_1 dw/w$. Its residue is P_1 , its horizontal monodromy is $\exp(-2\pi i P_1) = I$, and the map α is a morphism to $\mathcal{K}_{\text{ord}} = [E\mathcal{O}_X \xrightarrow{d} E\Omega_X^1(\log D)]$.

Here is the complete local calculation of the extension at infinity. On analytic power series the E_0 differential, with dw/w as the one-form frame, is $\sum a_n w^n \mapsto \sum n a_n w^n$. Its kernel and cokernel are the constant terms, both E_0 . The E_1 differential is

$$\sum_{n \geq 0} b_n w^n \mapsto \sum_{n \geq 0} (n+1) b_n w^n. \quad (\text{DS20})$$

It is invertible: its inverse divides the n th coefficient by $n+1$, which preserves every disk of convergence. Thus the E_1 part has zero derived stalk at infinity. On a punctured disk, integrating a Laurent series term by term divides its w^n coefficient by n for $n \neq 0$; the only one-form class remaining is dw/w with integral $2\pi i$ around a positively oriented w -circle. These statements, together with the usual local primitive on a disk, identify the E_0 part with $Rj_* \underline{E}_0$ and the E_1 part with $j_! \underline{E}_1$, where $j : X \setminus D \hookrightarrow X$. The identification for $j_!$ is the extension-by-zero map of flat sections; its stalk cohomology is an isomorphism on the open disk and is zero on both sides at infinity by (DS20). The identification for Rj_* is the restriction map to the punctured disk and the two Laurent calculations. This constructs the comparison on the present curve rather than assigning compact support by name.

Its entire global cohomology is

$$\mathbb{H}^0(X, \mathcal{K}) = E_0, \quad \mathbb{H}^1(X, \mathcal{K}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}) = E_1 \otimes \mathbb{T}, \quad \mathbb{T} = H^1(X, \Omega_X^1). \quad (\text{DS21})$$

For a direct verification, $\Omega_X^1(\log D) = \mathcal{O}_X(-1)$. The two terms of $\mathcal{K}$ are therefore $E_0\mathcal{O} \oplus E_1\mathcal{O}(-1)$ and $E_0\mathcal{O}(-1) \oplus E_1\mathcal{O}(-2)$. Using the two affine charts, polynomial division of Laurent polynomials proves

$$\begin{aligned} H^0(\mathcal{O}) &= \mathbb{C}, & H^1(\mathcal{O}) &= 0, & H^0(\mathcal{O}(-1)) &= H^1(\mathcal{O}(-1)) = 0, \\ H^0(\mathcal{O}(-2)) &= 0, & H^1(\mathcal{O}(-2)) &= \mathbb{C}. \end{aligned} \quad (\text{DS22})$$

More explicitly, after identifying $\mathcal{O}(-2)$ with Ω_X^1 , a Laurent one-form on the overlap has every term except dw/w removable by a one-form on one of the charts. The coefficient of dw/w is not removable and is its stated one-dimensional quotient. Retain $\mathbb{T}$ as this quotient line, with its original generator $[dz/z] = -[dw/w]$. The double complex for these two terms has only the entries E_0 in total degree zero and $E_1 \otimes \mathbb{T}$ in total degree two. There is no differential to or from them, proving (DS21) without an uncomputed higher differential.

Taking the quotient of the two complexes in (DS19) gives

$$\mathcal{Q}_D = [i_* E_1 \xrightarrow{0} i_* E_1 dw/w] \quad \text{in degrees } 0, 1. \quad (\text{DS23})$$

The zero differential follows by reducing $w\partial_w$ modulo w . The termwise short exact sequence $0 \rightarrow \mathcal{K} \rightarrow \mathcal{K}_{\text{ord}} \rightarrow \mathcal{Q}_D \rightarrow 0$ and the fact $\mathbb{H}^0(\mathcal{K}_{\text{ord}}) = E$, $\mathbb{H}^{>0}(\mathcal{K}_{\text{ord}}) = 0$ give

$$0 \rightarrow E_0 \rightarrow E \rightarrow E_1 \rightarrow 0, \quad E_1 dw/w \xrightarrow{\delta} E_1 \otimes \mathbb{T} \quad \text{an isomorphism.} \quad (\text{DS24})$$

Fix the Čech convention $s_V - s_U$ for the overlap, with U the finite chart and V the infinity chart, and the total-complex differential $\delta + (-1)^p \nabla$ on Čech degree p . Lifting a boundary one-form as $(0, v dw/w)$ makes its connecting class $v[dw/w]$; thus $\delta(v dw/w) = v \otimes [dw/w] = -v \otimes [dz/z]$. In the original finite coordinate $dw/w = -dz/z$. Both that sign and the chosen positive orientation of the w -circle are retained. Under (DS21), the ordinary comparison image is exactly E_0 in degree zero and zero in degrees one and two. The full E_1 occurs in its boundary comparison and in degree two.

1.5 The divisor twist in the support-compatible dual

The bundle dual in (DS18) and the dual de Rham comparison are joined by the exact additional map of line bundles $\mathcal{O}(-D) \hookrightarrow \mathcal{O}$. Define

$$\mathcal{E}^{\text{cd}} = \mathcal{E}^\vee \otimes \mathcal{O}(-D), \quad \mathcal{K}^{\text{cd}} = [\mathcal{E}^{\text{cd}} \xrightarrow{\nabla^{\text{cd}}} \mathcal{E}^{\text{cd}} \otimes \Omega_X^1(\log D)]. \quad (\text{DS25})$$

The dual connection has residue $-P_1^\vee$. Multiplication of its local frame by w , the local generator of $\mathcal{O}(-D)$, adds I to the residue. Hence the residue of (DS25) is $P_0^\vee$, and

$$\mathcal{E}^{\text{cd}} = E_0^* \mathcal{O}(-D) \oplus E_1^* \mathcal{O}, \quad \frac{m}{\mathbb{H}^m(\mathcal{K}^{\text{cd}})} \left| \begin{array}{ccc} 0 & 1 & 2 \\ E_1^* & 0 & E_0^* \otimes \mathbb{T} \end{array} \right. \quad (\text{DS26})$$

The proof is exactly the power-series calculation (DS20) and the four line-bundle groups (DS22), now with the two summands exchanged.

Evaluation is a morphism $\mathcal{E} \otimes \mathcal{E}^{\text{cd}} \rightarrow \mathcal{O}(-D)$. The connections obey the Leibniz identity for evaluation, since the dual connection cancels the original connection term. Wedge product therefore gives a cochain map

$$\mathcal{K} \otimes_{\mathbb{C}} \mathcal{K}^{\text{cd}} \longrightarrow [\mathcal{O}(-D) \xrightarrow{d} \Omega_X^1] =: \mathcal{K}_c(\mathbb{C}). \quad (\text{DS27})$$

On a degree-zero tensor it is evaluation; on degree-one tensors it is evaluation times the retained one-form; on two one-forms it is zero because $\Omega_X^2 = 0$. These prescriptions and the tensor sign in (DS9) verify the cochain identity in every degree. The degree-wise evaluation and wedge maps factor through their $\mathcal{O}_X$ -balanced tensor products; the displayed complex tensor product is over $\mathbb{C}$, so its differential is defined. Define the trace on $\mathbb{H}^2(\mathcal{K}_c(\mathbb{C}))$ by

$$\text{Tr}_D([dz/z]) = 1, \quad \text{Tr}_D([dw/w]) = -1 = -\frac{1}{2\pi i} \int_{|w|=\epsilon}^{\text{counterclockwise}} \frac{dw}{w}. \quad (\text{DS28})$$

Holomorphic one-forms on either chart have zero overlap residue, so this is well-defined on (DS22). The induced pairings are

$$E_0 \times E_0^* \rightarrow \mathbb{C}, \quad E_1 \times E_1^* \rightarrow \mathbb{C}, \quad (\text{DS29})$$

ordinary evaluation after applying the retained trace on $\mathbb{T}$, in complementary degrees 0, 2 and 2, 0. They are perfect, since a nonzero vector is detected by a linear functional. Thus (DS25)–(DS29) prove the support-compatible duality, including the dimension-one degree change and its actual boundary line factor. The finite-field analogue of the trace target is $\overline{\mathbb{Q}}_\ell(-1)$; the characteristic-zero calculation here has the explicit residue normalization (DS28).

1.6 Retaining the arithmetic action through the exceptional fibre

Transporting the original A through α gives

$$\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}. \quad (\text{DS30})$$

It extends regularly on $\mathcal{E}$ exactly when $AE_0 \subseteq E_0$, since its sole possible negative power is $w^{-1}P_1AP_0$. The explicit repair on the same ambient arithmetic target is

$$F_A = \sum_{a=0}^{q-1} A^a E_0, \quad \mathcal{M}_A = F_A \mathcal{O}_X + E \mathcal{O}_X(-D). \quad (\text{DS31})$$

Cayley-Hamilton puts $A^q E_0$ in the displayed sum, proving invariance; every invariant subspace containing E_0 contains this sum. Similarly every coherent A -stable submodule of $E\mathcal{O}$ containing $\mathcal{M}$ contains $\sum_{a=0}^{q-1} A^a \mathcal{M}$, which is exactly $\mathcal{M}_A$ in (DS31). There are exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_*(F_A/E_0) \rightarrow 0, \\ 0 \rightarrow \mathcal{M}_A \rightarrow E\mathcal{O}_X \rightarrow i_*(E/F_A) \rightarrow 0. \end{aligned} \quad (\text{DS32})$$

The second is A equivariant on every term. The connection d on $E\mathcal{O}$ preserves $\mathcal{M}_A$ logarithmically, because $d(wv) = w dv + v dw$ lies in $\mathcal{M}_A \otimes \Omega^1(\log D)$. It commutes with the constant original A . Define its two-term complex $\mathcal{K}_A$ with the same degree convention as (DS19). Calculation with any vector-space complement of F_A proves, intrinsically,

$$\mathbb{H}^0(\mathcal{K}_A) = F_A, \quad \mathbb{H}^1(\mathcal{K}_A) = 0, \quad \mathbb{H}^2(\mathcal{K}_A) = (E/F_A) \otimes \mathbb{T}, \quad (\text{DS33})$$

with actions $A|_{F_A}$ and $\bar{A}$, respectively. The identifications are independent of that auxiliary complement: the degree-zero map is inclusion into E and the degree-two map is the connecting quotient map in (DS32), as in (DS24).

Its dual lattice has a coordinate-free formula. Put $F_A^\perp = \{\lambda \in E^* : \lambda(F_A) = 0\}$. Then

$$\mathcal{M}_A^\vee(-D) = F_A^\perp \mathcal{O}_X + E^* \mathcal{O}_X(-D). \quad (\text{DS34})$$

Indeed a meromorphic functional belongs to the left side precisely when its value on every element of $F_A \mathcal{O} + wE\mathcal{O}$ lies in $w\mathcal{O}$. This first forces it to be holomorphic, by testing wE , and then forces its constant term to annihilate F_A . This is exactly the right side. Fix the additive dual action as $-A^\vee$, so evaluation has the zero action on its one-dimensional target. The displayed lattice is invariant under this action because F_A is invariant. Equations (DS27)–(DS29) therefore become the perfect equivariant pairings $F_A \times F_A^*$ and $(E/F_A) \times F_A^\perp$. For the original real integer k , replacing the dual action by $kI - A^\vee$ puts the scalar k action on the evaluation target, as checked in (DS6). The passage to the metric antidual is the specified conjugate-linear map $\mathfrak{c} : E^\vee \rightarrow E^\vee$, $\mathfrak{c}(\lambda)(v) = \bar{\lambda}(v)$. In the fixed polynomial basis it conjugates the functional's coefficient column, and hence $\mathfrak{c}(kI - A^T) = (kI - A^*)\mathfrak{c}$. This proves the exact coefficient map between the linear dual and the original metric antidual convention. For every polynomial p , an adapted basis to F_A proves

$$\text{Tr } p(A) = \text{Tr } p(A|_{F_A}) + \text{Tr } p(\bar{A}). \quad (\text{DS35})$$

This retains the complete arithmetic trace across the calculated cohomological degrees. There is no need to presume that the metric kernel E_0 was an arithmetic constituent.

The source cost of (DS31) remains visible. If $s_A = \dim F_A - \dim E_0$, choose a basis first for E_0 , then for F_A/E_0 , and then for E/F_A . The local inclusion $\mathcal{M} \rightarrow \mathcal{M}_A$ has diagonal blocks I, wI_{s_A}, I and determinant w^{s_A} . If the metric on $\mathcal{M}_A$ is transported on the punctured chart from the original source metric on $\mathcal{M}$, the determinant identity is

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w). \quad (\text{DS36})$$

It follows by taking determinants of $G_{\mathcal{M}} = J^* G_A J$. Thus the action-stable complex (DS33) retains the exceptional contribution through an explicitly measured metric pole.

1.7 The same curvature, its source map, and the weight interface

For $\mathcal{L} = (\det \mathcal{E})^\vee$ with its original source metric, put

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i. \quad (\text{DS37})$$

The second equality is $K_j G_i - I = (K_j - K_i)G_i$. The first expression proves self-adjointness and positivity in G_j ; the second proves the same in G_i , because $G_i Z = G_i(K_j - K_i)G_i \succeq 0$ and $K_j - K_i \succeq 0$. No zero eigenvalue is removed. Differentiating $\log \det(G_j + |z|^2 C)$ yields

$$c_1(\mathcal{L}) = \frac{i}{2\pi} \kappa(|z|^2) dz \wedge d\bar{z}, \quad \kappa(t) = \text{Tr}(Z(I + tZ)^{-2}). \quad (\text{DS38})$$

To verify every coefficient, set $f(t) = \log \det(I + tZ)$. Then $f'(t) = \text{Tr}(Z(I + tZ)^{-1})$ and $f'(t) + t f''(t) = \kappa(t)$. Since $\partial \bar{\partial} f(|z|^2) = (f'(|z|^2) + |z|^2 f''(|z|^2)) dz \wedge d\bar{z}$, this proves (DS38) with the stated sign. For a positive eigenvalue λ , $\int_0^\infty \lambda(1 + t\lambda)^{-2} dt = 1$; for a zero eigenvalue the integrand is zero. Since $i dz \wedge d\bar{z} = 2dx \wedge dy$, polar integration gives the degree and the logarithmic moment

$$\int_X c_1(\mathcal{L}) = r, \quad \log \frac{\det G_i}{\det G_j} = \int_0^1 \log(1/t) \kappa(t) dt. \quad (\text{DS39})$$

For the second equality, write $M(t) = t f'(t)$, so $M' = \kappa$ and $M(0) = 0$. Nonnegative Fubini gives $\int_0^1 \log(1/t) \kappa(t) dt = \int_0^1 t^{-1} M(t) dt = f(1) - f(0)$.

The full source observation is also retained. With

$$P_z = R(z)G(z)^{-1}R(z)^*O_j, \quad \mathcal{N}_z = (I - P_z)\Delta, \quad G(z) = G_j + |z|^2 C, \quad (\text{DS40})$$

orthogonal projection gives

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C, \quad \text{Tr}(G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z) = \kappa(|z|^2). \quad (\text{DS41})$$

Indeed $R(z)^* O_j \Delta = \bar{z} C$; substitute into $\Delta^* O_j (I - P_z) \Delta$. Factoring $G(z) = G_j(I + |z|^2 Z)$ in the trace proves the second equality. Applying the actual remainder map gives

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C, \quad B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C. \quad (\text{DS42})$$

Their sum is the original theta relation. In the split lift its quotient value is the supported zero e , with both prequotient pieces still given by (DS42).

The curvature jets are the retained relation moments:

$$\kappa^{(n)}(0) = (-1)^n (n+1)! \text{Tr}(Z^{n+1}), \quad t_1 = \kappa(0), \quad t_2 = -\frac{1}{2} \kappa'(0), \quad \mathcal{A}_2 = \frac{1}{2} (t_1^2 - t_2). \quad (\text{DS43})$$

These follow by differentiating each rational eigenvalue term in (DS38); finiteness of the matrix makes the differentiation exact. The dimension-one divisor degree in (DS39), the arithmetic trace in (DS35), and the source metric pole in (DS36) are now related by specified maps on the same compactified curve.

For completeness, exponentiation supplies the precise multiplicative observer of the original additive action, for every real t :

$$F_t = \exp(tA) : E \rightarrow E, \quad F_t|_{F_A} = \exp(tA|_{F_A}), \quad \bar{F}_t = \exp(t\bar{A}). \quad (\text{DS44})$$

On the full generalized eigenspace of A at ρ , write $A = \rho I + N$, $N^m = 0$ with its actual order. Then $F_t = e^{t\rho} \sum_{a=0}^{m-1} t^a N^a / a!$; in particular it retains the nilpotent terms and has eigenvalue modulus $e^{t \text{Re } \rho}$. Choosing $t = \log Q_0$ gives the exact number $Q_0^{\text{Re } \rho}$. This is the proved operator map connecting the additive spectral parameter and a multiplicative weight observer. It does not provide a finite-field descent or an étale Frobenius structure. Those structures in (DS7) have the specific types $X_0/\mathbb{F}_{Q_0}$, a smooth $\mathbb{Q}_\ell$ -sheaf and its geometric Frobenius action; the constructed characteristic-zero object has

instead the fully calculated complexes (DS19), (DS25), (DS33) and the commuting action (DS44). Its complete comparison to those source operations is (DS1)–(DS9), (DS24), (DS27), and (DS34), with their actual kernels.

Finally the original positive metric supplies its precise additive weight measurement on every invariant inclusion $I_F : F \hookrightarrow E$:

$$G_F = I_F^* G(z) I_F, \quad W_F = A_F^* G_F + G_F A_F - k G_F, \quad \text{Tr}(G_F^{-1} W_F) = 2 \text{Re Tr } A_F - k \dim F. \quad (\text{DS45})$$

The identity follows by cyclicity of the finite trace, with $A I_F = I_F A_F$. Thus k remains the original tensor parameter, and all constituents of E are retained through (DS32)–(DS35). Equations (DS38)–(DS43) make the current endpoint estimate an estimate on a specified source-induced curvature measure. Its total degree alone does not bound its logarithmic moment: for the explicitly declared matrix family $G_j = I_r$, $C = L I_r$, $L > 0$, these quantities are respectively r and $r \log(1 + L)$, and the mass on $0 \leq t \leq 1/L$ is exactly $r/2$ by direct integration. The original arithmetic source, its relation map (DS13), its full dual comparison (DS25)–(DS34), and the actual moment observers (DS43) therefore remain the concrete data for the continuing curvature calculation.

1.8 Every divisor depth and the unchanged conormal directions

For a specified A -invariant subspace $F \subseteq E$ and an integer $m \geq 1$, retain

$$\mathcal{M}_{F,m} = F \mathcal{O}_X + E \mathcal{O}_X(-mD), \quad \mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D)]. \quad (\text{DS46})$$

The local coefficient differential on $w^m(E/F)$ is $w \partial_w + m$. The inclusion $\mathcal{K}_{F,m} \hookrightarrow \mathcal{K}_{F,1}$ has quotient supported at D , with coefficient basis $w, w^2, \dots, w^{m-1}$ in each of degrees zero and one and differential

$$w^a v \longmapsto a w^a v dw/w \quad (1 \leq a < m). \quad (\text{DS47})$$

Define $h(w^a v dw/w) = a^{-1} w^a v$ on degree one and $h = 0$ on degree zero. Then $dh + hd = I$ on each displayed basis element, so this entire quotient is contractible. Thus the inclusion is a quasi-isomorphism, equivariant for the original A , and

$$\mathbb{H}^0(\mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(\mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(\mathcal{K}_{F,m}) = (E/F) \otimes \mathbb{T}. \quad (\text{DS48})$$

This preserves every higher coefficient until its explicit differential (DS47) contracts it. In original arithmetic degree k the shifted complex is $\mathcal{K}_{F,m}[-k]$, so the two surviving degrees are $k, k+2$ and

$$\text{Str}(p(A) \mid R\Gamma \mathcal{K}_{F,m}[-k]) = (-1)^k \text{Tr}_E p(A). \quad (\text{DS49})$$

The equality follows from (DS35); multiplication by $p(A)$ acts trivially on the factor $\mathbb{T}$.

The exact high-depth dual is

$$\mathcal{M}_{F,m}^\vee(-D) = F^{\text{ann}} \mathcal{O}_X((m-1)D) + E^* \mathcal{O}_X(-D) \subset E^* \otimes \mathcal{O}_X(*D). \quad (\text{DS50})$$

At infinity, testing a meromorphic functional on $w^m E$ permits poles of order at most $m-1$, and testing on F forces every coefficient of powers $w^{1-m}, \dots, w^0$ to annihilate F . Its remaining coefficients are $w E^* \mathcal{O}$. This proves (DS50), including its ambient meromorphic sheaf. In a coefficient complement its residues are 1 on F^* and $1-m$ on $(E/F)^*$. The natural inclusion of the depth-one dual into (DS50) has quotient basis $w^{1-m}, \dots, w^{-1}$ in both degrees, with differential $w^a v \mapsto a w^a v dw/w$ and inverse a^{-1} , so it is contractible. Therefore its dual cohomology is F^{ann} in degree zero and $F^* \otimes \mathbb{T}$ in degree two. Evaluation into $\mathcal{O}(-D)$ and (DS27) give the same perfect complementary-degree pairings. All negative powers, residues and the divisor twist have been retained.

The independent conormal depth has the explicit map

$$\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^m) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/(w^m)) \otimes (\mathcal{P}/I^a), \quad [f(w)P] \mapsto [f] \otimes [P], \quad (\text{DS51})$$

where $\mathcal{P} = \mathbb{C}[s_1, \dots, s_k]$ and $I = (h(s_1), \dots, h(s_k))$. Reduction of the w -degree writes the left side uniquely as $\sum_{b=0}^{m-1} w^b [P_b]_{I^a}$. This supplies the inverse and proves injectivity as well as surjectivity. Restriction in either depth commutes with the other. Differentiation in w has the well-defined type from depth $m+1$ to depth m ; differentiation in s_j has type from I^{a+1} quotients to I^a quotients, because $\partial_{s_j} I^{a+1} \subset I^a$. On pure tensors the two derivatives commute, hence on their full spans. Lifting these maps by (DS10) retains the same support label and the receiving supported zero in both depth directions.

Pullback of (DS38) under the actual map $z \mapsto z^m$ has radial density

$$\kappa_m(t) = m^2 t^{m-1} \kappa(t^m), \quad t = |z|^2. \quad (\text{DS52})$$

Indeed differentiate $f(t^m)$ twice in the combination $f'_m + t f''_m$ used in (DS38). Substitution $s = t^m$ proves $\int_0^\infty \kappa_m(t) dt = mr$ and $\int_0^1 \log(1/t) \kappa_m(t) dt = \int_0^1 \log(1/s) \kappa(s) ds$. Thus increasing divisor depth multiplies degree by m while leaving the original endpoint logarithmic moment fixed.

1.9 The invariant quotient gives an exact coupling subtraction

The following quantitative calculation is the delivered logarithmic constituent-control result, now proved in the same source notation. Take $I : F \hookrightarrow E$ and $\pi : E \twoheadrightarrow E/F$ with $AI = IA_F$ and $\pi A = A_Q \pi$, and put $G(t) = G_0 + tC$, $G_0 = G_j$, $0 \leq t \leq 1$. The cases $F = 0, E$ use the empty determinant 1, empty trace 0, and the unique maps on the zero-dimensional space; all formulas below still hold. For any initial coefficient lift $\bar{H}$, define

$$H_0 = \bar{H} - I(I^* G_0 I)^{-1} I^* G_0 \bar{H}, \quad \pi H_0 = I_{E/F}, \quad I^* G_0 H_0 = 0. \quad (\text{DS53})$$

These last two identities follow from $\pi I = 0$ and matrix multiplication. Changing $\bar{H}$ by IU cancels the change in (DS53); thus the orthogonal lift itself is independent of the initial lift. In the original bases define

$$\begin{aligned} G_{F,0} &= I^* G_0 I, \quad C_F = I^* C I, \quad D_F = I^* C H_0, \\ Q_0 &= H_0^* G_0 H_0, \quad C_Q = H_0^* C H_0, \quad G_F(t) = G_{F,0} + t C_F, \\ H_t &= H_0 - t I G_F(t)^{-1} D_F. \end{aligned} \quad (\text{DS54})$$

Here Q_0 is the original quotient Gram, not the finite-field cardinality used in (DS7). Every inverse is defined: I and H_0 are injective on their respective coefficient spaces, and $G(t) > 0$. In coordinates $[I, H_0]$ the original metric is exactly

$$\begin{pmatrix} G_F(t) & t D_F \\ t D_F^* & Q_0 + t C_Q \end{pmatrix}. \quad (\text{DS55})$$

The replacement $H_0 \mapsto H_t$ is the upper triangular coefficient change $\begin{pmatrix} I & -t G_F(t)^{-1} D_F \\ 0 & I \end{pmatrix}$ of determinant 1. It gives

$$I^* G(t) H_t = 0, \quad \pi H_t = I_{E/F}, \quad Q(t) := H_t^* G(t) H_t = Q_0 + t C_Q - t^2 D_F^* G_F(t)^{-1} D_F > 0. \quad (\text{DS56})$$

Expansion proves the formula, and injectivity of H_t proves positivity. It is also the quotient norm: any other lift of v is $H_t v + Ix$, whose squared norm is $v^* Q(t) v + x^* G_F(t) x$ by orthogonality. The unique minimum is therefore attained at $H_t v$.

Define the actual coupling map

$$\beta_t = G_F(t)^{-1} I^* C H_t : E/F \rightarrow F. \quad (\text{DS57})$$

Differentiate $\pi H_t = I$ to write $H'_t = I U_t$; differentiating $I^* G(t) H_t = 0$ then gives $I^* C H_t + G_F(t) U_t = 0$. Therefore $H'_t = -I \beta_t$. In the derivative of $Q(t) = H_t^* G(t) H_t$ the two terms containing H'_t vanish by orthogonality. Differentiating the remaining term gives

$$Q'(t) = H_t^* C H_t, \quad Q''(t) = -2 \beta_t^* G_F(t) \beta_t. \quad (\text{DS58})$$

The factor two follows because $I^*CH_t = G_F(t)\beta_t$ and its adjoint supply the two equal terms. The exact integral remainder for $Q(1)$ is $Q(1) = Q(0) + Q'(0) + \int_0^1 (1-t)Q''(t)dt$; insert (DS56) at the two endpoints to get

$$\Pi_F := D_F^* G_F(1)^{-1} D_F = 2 \int_0^1 (1-t) \beta_t^* G_F(t) \beta_t dt. \quad (\text{DS59})$$

This identifies the finite subtraction with an integrated squared map on the original quotient source. The action defect is the unique map $a_t : E/F \rightarrow F$ with

$$AH_t - H_t A_Q = I a_t, \quad a'_t = \beta_t A_Q - A_F \beta_t. \quad (\text{DS60})$$

The first left side has zero image under π , so it lies in $\text{im } I$, which proves existence and uniqueness. Differentiating and inserting $H'_t = -I\beta_t$ proves the second formula, using injectivity of I . Thus the nilpotent extension data are retained in a_t even when the two constituent actions have equal eigenvalues.

For clarity the entire Sylvester operator has a finite exact description. On a pair of primary spaces let $A_F = \lambda I + N_F$, $A_Q = \mu I + N_Q$, with $N_F^a = 0$ and $N_Q^b = 0$. Put $\mathcal{N}(X) = XN_Q - N_F X$. The two multiplications commute, so the binomial expansion makes $\mathcal{N}^{a+b-1} = 0$. If $\mu \neq \lambda$, the inverse of $X \mapsto XA_Q - A_F X$ is

$$Y \mapsto \sum_{r=0}^{a+b-2} \frac{(-1)^r}{(\mu - \lambda)^{r+1}} \mathcal{N}^r(Y). \quad (\text{DS61})$$

Multiplying the finite geometric series by $(\mu - \lambda)I + \mathcal{N}$ leaves identity and a zero terminal power. At $\mu = \lambda$, the operator is exactly $\mathcal{N}$; its kernel is the space $\{X : XN_Q = N_F X\}$ and its image is the span of the explicit matrices $E_{uv}N_Q - N_F E_{uv}$. Matrix-unit expansion proves that this span is its whole image. This keeps the kernel and every coefficient of the image without a nonexistent inverse.

The same original control $W(t) = A^*G(t) + G(t)A - kG(t)$ obeys

$$\begin{aligned} I^*W(t)I &= A_F^*G_F(t) + G_F(t)A_F - kG_F(t), \\ H_t^*W(t)H_t &= A_Q^*Q(t) + Q(t)A_Q - kQ(t). \end{aligned} \quad (\text{DS62})$$

The first identity uses invariance of F . For the second insert (DS60); both terms containing Ia_t vanish by (DS56). On the original theta source write $r(z) = \mathcal{V}R(z)$ and retain its actual primitive $D^{(k)}r(z) - r(z)A = dK(z)$, with $K(z) = (1-z)K_j + zK_i$ in the inherited primitive notation. Then exact multiplication gives

$$D^{(k)}r(z)H_t - r(z)H_t A_Q = dK(z)H_t + r(z)Ia_t, \quad t = |z|^2. \quad (\text{DS63})$$

Here $D^{(k)}$ differentiates the original source variables, not the comparison parameter z or t . The first term is killed by the original theta quotient. The second has its retained image in the arithmetic constituent F and is killed by the further quotient $E \rightarrow E/F$. Formula (DS63) proves the precise map between these two successive supported-zero operations.

Put $Q_+ = Q_0 + C_Q$, $\Xi_F = Q_+^{-1}\Pi_F$. Then Ξ_F is self-adjoint and nonnegative for Q_+ , and $Q(1) = Q_+ - \Pi_F > 0$ proves that all its eigenvalues belong to $[0, 1)$. Taking determinants in (DS55)–(DS56), with the fixed coefficient determinant $|\det[I, H_0]|^2$ cancelled only between the same two original endpoints, gives

$$\begin{aligned} \log \frac{V_i}{V_j} &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F) \\ &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}. \end{aligned} \quad (\text{DS64})$$

For each eigenvalue $0 \leq x < 1$, integrating the geometric series $\sum_{a \geq 0} x^a = (1-x)^{-1}$ from zero to x proves the logarithm series. There are finitely many eigenvalues, so their sum yields the second

equality and every term is nonnegative. Its first p terms therefore give the stated upper estimate, and these estimates decrease to the exact original endpoint volume.

There is also an explicit remainder from the same matrices. If $\rho = \lambda_{\max}(\Xi_F) < 1$, then for every $p \geq 0$,

$$0 \leq \sum_{r=p+1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r} \leq \frac{\text{Tr}(\Xi_F^{p+1})}{(p+1)(1-\rho)}. \quad (\text{DS65})$$

Indeed $\text{Tr}(\Xi_F^{p+1+a}) \leq \rho^a \text{Tr}(\Xi_F^{p+1})$ on each nonnegative eigenvalue, and $p+1+a \geq p+1$; summing the resulting geometric series proves the inequality. The value ρ is the largest generalized eigenvalue of (Π_F, Q_+) of this same source pair, with no independently assumed uniform gap. On the zero-dimensional quotient $E/F = 0$, define $\rho = 0$; every trace and both sides of (DS65) are then zero, so the same formula includes that endpoint.

The two positive terms in (DS64) are the logarithmic determinants of the compressed positive operators $Z_F = G_{F,0}^{-1}C_F$ and $Z_Q = Q_0^{-1}C_Q$. Consequently the already proved two-trace majorant, applied to each of these specified matrices in its own full dimension, can be combined with every finite subtraction in (DS64). This integration retains the extension map (DS60), the quotient source (DS63), and the original total curvature (DS39) together; the cohomological constituent decomposition does not remove its measurable coupling.

1.10 The degree-one exponential family joins the same coefficient object

The delivered exponential-comparison note constructs the additional degree-one family from the unchanged polynomial $\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a$. Its immediate join to the present cohomology calculation is as follows. Use new independent parameters u, t and retain every coefficient of

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS, \quad R = \mathbb{C}[u, t]. \quad (\text{DS66})$$

The two-term coefficient complex is

$$\mathcal{C} = [R[S] \xrightarrow{\mathcal{D}} R[S]dS], \quad \mathcal{D}P = (uP' + (\chi - t)P)dS. \quad (\text{DS67})$$

For a nonzero polynomial P of degree a and leading coefficient $b \in R$, $\mathcal{D}P/dS$ has degree $a+q$ and leading coefficient b . Successively subtracting $\mathcal{D}(bS^a)/dS$ cancels each highest term in a polynomial of degree at least q and terminates. Uniqueness follows from the same degree calculation. Hence $R[S] = \text{im}(\mathcal{D}/dS) \oplus R[S]_{<q}$ as coefficient modules, $H^0(\mathcal{C}) = 0$, and $\mathcal{H} = H^1(\mathcal{C})$ is free with the original basis $[1], \dots, [S^{q-1}]$. At $u = t = 0$ the differential is multiplication by the literal χ , so $\mathcal{H}/(u, t)\mathcal{H} = E$ with every nilpotent jet.

The commutators are $[\mathcal{D}/dS, \partial_t] = I$ and $[\mathcal{D}/dS, -S/u] = -I$ after inverting u . They prove that $\nabla_t = \partial_t - S/u$ descends through the differential. Define $\mathcal{R} : E \rightarrow E$ by sending a remainder to its S^{q-1} coefficient times $[1]$. Reducing S^q with the relation $[\chi - t] = 0$ shows in the displayed basis that

$$\nabla_t = \partial_t - \frac{A + t\mathcal{R}}{u}, \quad (-u\nabla_t) \bmod (u, t) = A. \quad (\text{DS68})$$

The regular operator $-u\partial_t + A + t\mathcal{R}$ preserves $(u, t)\mathcal{H}$: on uv its derivative contribution is $-u^2\partial_tv$, and on tv it is $-uv - ut\partial_tv$. This verifies the asserted specialization as an actual induced operator, rather than specializing a singular connection symbol.

The logarithmic comparison retains the original exact coefficient sequence $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$, with the outer terms occurring in (DS48). The common full E is obtained through its actual outer comparison complexes: $\mathbb{H}^0(\mathcal{K}_{E,1}) = E$ and $\mathbb{H}^2(\mathcal{K}_{0,1}) = E \otimes \mathbb{T}$. The maps to and from the mixed complex are $0 \rightarrow F \rightarrow E$ and $E \otimes \mathbb{T} \rightarrow (E/F) \otimes \mathbb{T} \rightarrow 0$. At the exponential special fibre the cochain map $[\mathbb{C}[S] \xrightarrow{\chi} \mathbb{C}[S]dS \rightarrow E[-1]]$ is zero in degree zero and $PdS \mapsto [P]_{\chi}$ in

degree one. Its kernel complex is the isomorphism $\mathbb{C}[S] \xrightarrow{\chi} \chi \mathbb{C}[S] dS$ and is contractible by division by χ on its image. This proves the special-fibre quasi-isomorphism. The resulting map to the full coefficient object of the logarithmic outer complexes is the identity on $[1], \dots, [S^{q-1}]$, intertwining A . Composing with $E \rightarrow E\mathcal{O}_X, v \mapsto v \otimes 1$, gives the map into the original constant target of (DS32). It retains the coefficient extension and its map a_t in (DS60). The original theta comparison $j_E r_N = I$, $j_E D^{(k)}_t = A j_E$ makes this identity a source comparison as well:

$$D^{(k)}_t = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D^{(k)}_t = (A + t \mathcal{R}) j_E, \quad D^{(k)}_t r_N - r_N (A + t \mathcal{R}) = dK_N. \quad (\text{DS69})$$

Both equalities follow by multiplying and using $j_E r_N = I$; the added terms cancel exactly in the second equality. Thus the source primitive is unchanged and the full kernel $\ker j_E$ remains. The exponential note's contour periods and finite-field specializations develop this new family further; equations (DS66)–(DS69) are its complete coefficient, cohomological and source-level join to the logarithmic machinery in this sidebar.